

# Generation Interconnection System Impact Study Report

## For Queue Project Q20-1439

Southwest Research Institute 138 kV

May 01, 2021

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1439. The Interconnection Customer proposes a 54.9 MW battery energy storage facility interconnecting at the Southwest Research Institute 138 kV substation in the Virginia area. The request was received on March 29, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$30,524,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

For the loss of Clark County to Kearneyville 345 KV, the Arvin to Shuffleton 138 KV line loads to 135.3% of its 223 MVA rating. Project DFAX: 55.6%.

For the loss of Douglas City to Southwest Research Institute 138 KV Line, the Garden Corners to Arvin 138 KV L loads to 126.6% of its 171 MVA rating. Project DFAX: 18.9%.

The Arvin to Southwest Research Institute 138 KV L is overloaded to 122.6% of its 226 MVA emergency rating for the loss of the Kearneyville to Clark County 345 KV. The project contributes 47.5% DFAX (26.1 MW) to this overload.

Palmyra to Grimmway-Malaga 138 KV L: 121.1% loading (313 MVA) following the outage of the Clark County to Kearneyville 345 KV; DFAX = 16.6%.

The Palmyra to Glenbrook 138 KV is overloaded to 120.6% of its 325 MVA emergency rating for the loss of the Clark County to Kearneyville 345 KV. The project contributes 16.6% DFAX (9.1 MW) to this overload.

Redman to Glenbrook 138 KV line: 120.1% loading (336 MVA) following the outage of the Clark County Substation to Kearneyville 345 KV line; DFAX = 16.6%.

For the loss of Clark County to Kearneyville 345 KV, the Douglas City to Southwest Research Institute 138 KV loads to 120.0% of its 261 MVA rating. Project DFAX: 52.5%.

For the loss of Clark County Substation to Kearneyville 345 KV L, the Reman to Gallegos 138 KV loads to 119.8% of its 344 MVA rating. Project DFAX: 16.6%.

Garden Corners to Douglas City 138 KV: 119.7% loading (214 MVA) following the outage of the Douglas City to Southwest Research Institute 138 KV L; DFAX = 18.9%.

The Killingsworth to Gallegos 138 KV is overloaded to 119.5% of its 354 MVA emergency rating for the loss of the Clark County to Kearneyville 345 KV. The project contributes 16.6% DFAX (9.1 MW) to this overload.

The Douglas City to Millville 138 KV line is overloaded to 115.2% of its 461 MVA emergency rating for the loss of the Kearneyville to Clark County Substation 345 KV. The project contributes 33.6% DFAX (18.5 MW) to this overload.

Millville to Sleepy Hollow 138 KV: 114.9% loading (474 MVA) following the outage of the Clark County to Kearneyville 345 KV L; DFAX = 33.6%.

For the loss of Clark County to Kearneyville 345 KV, the Grimmway-Malaga to Sleepy Hollow 138 KV loads to 114.7% of its 487 MVA rating. Project DFAX: 33.6%.

Shuffleton to Claytonia 138 KV line: 112.7% loading (241 MVA) following the outage of the Clark County Substation to Keaneyville 345 KV; DFAX = 55.6%.

For the loss of Tiger Bay to Dranesville 345 KV, the Trumann to Rossmoor South 138 KV Line loads to 111.3% of its 405 MVA rating. Project DFAX: 5.2%.

Rossmoor S to Tramway 138 KV: 108.5% loading (468 MVA) following the outage of the Dranesville to Tiger Bay 345 KV; DFAX = 5.2%.

For the loss of American to Urban Substation 345 KV, the Rio Dell to St. John 138 KV line loads to 107.3% of its 513 MVA rating. Project DFAX: 7.2%.

For the loss of Gallegos to Killingsworth 138 KV L, the Grimmway-Malaga to Gene 138 KV L loads to 107.3% of its 314 MVA rating. Project DFAX: 28.3%.

Gene to Tanner 138 KV: 107.2% loading (327 MVA) following the outage of the Killingsworth to Gallegos 138 KV; DFAX = 28.3%.

Smallwood to Killingsworth 138 KV: 107.2% loading (221 MVA) following the outage of the Kearneyville Substation to Clark County Substation 345 KV; DFAX = 8.6%.

The Tanner to LcecNo. 5 Substation 138 KV is overloaded to 106.8% of its 336 MVA emergency rating for the loss of the Gallegos to Killingsworth 138 KV L. The project contributes 28.3% DFAX (15.5 MW) to this overload.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                            | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Arvin to Shuffleton 138 KV line                     | Rebuild line with higher-capacity conductor | \$2,938,000   | 42% (\$1,239,000)     |
| Garden Corners to Arvin 138 KV L                    | Rebuild line with higher-capacity conductor | \$11,067,000  | 59% (\$6,542,000)     |
| Arvin to Southwest Research Institute 138 KV L      | Rebuild line with higher-capacity conductor | \$7,291,000   | 51% (\$3,734,000)     |
| Palmyra to Grimmway-Malaga 138 KV L                 | Rebuild line with higher-capacity conductor | \$1,959,000   | 19% (\$364,000)       |
| Glenbrook Substation to Palmyra 138 KV              | Rebuild line with higher-capacity conductor | \$2,405,000   | 19% (\$447,000)       |
| Redman to Glenbrook 138 KV Line                     | Rebuild line with higher-capacity conductor | \$1,577,000   | 19% (\$293,000)       |
| Douglas City to Southwest Research Institute 138 KV | Rebuild line with higher-capacity conductor | \$2,772,000   | 54% (\$1,488,000)     |
| Gallegos to Redman 138 KV                           | Rebuild line with higher-capacity conductor | \$2,243,000   | 19% (\$417,000)       |
| Garden Corners to Douglas City 138 KV               | Rebuild line with higher-capacity conductor | \$1,658,000   | 59% (\$980,000)       |
| Killingsworth to Gallegos 138 KV                    | Rebuild line with higher-capacity conductor | \$3,139,000   | 19% (\$583,000)       |
| Douglas City to Millville 138 KV L                  | Rebuild line with higher-capacity conductor | \$18,316,000  | 27% (\$4,939,000)     |

| Facility                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------|---------------------------------------------|---------------|-----------------------|
| Millville to Sleepy Hollow 138 KV line  | Rebuild line with higher-capacity conductor | \$9,146,000   | 27% (\$2,467,000)     |
| Grimmway-Malaga to Sleepy Hollow 138 KV | Rebuild line with higher-capacity conductor | \$2,391,000   | 27% (\$645,000)       |
| Shuffleton to Claytonia 138 KV line     | Rebuild line with higher-capacity conductor | \$3,494,000   | 42% (\$1,474,000)     |
| Rossmoor S to Trumann Substation 138 KV | Rebuild line with higher-capacity conductor | \$6,025,000   | 4% (\$255,000)        |
| Rossmoor S to Tramway 138 KV            | Reconductor line                            | \$684,000     | 4% (\$29,000)         |
| Rio Dell to St. John 138 KV line        | Reconductor line                            | \$1,571,000   | 12% (\$184,000)       |
| Gene to Grimmway-Malaga 138 KV          | Reconductor line                            | \$3,120,000   | 65% (\$2,027,000)     |
| Gene to Tanner 138 KV                   | Reconductor line                            | \$252,000     | 65% (\$164,000)       |
| Killingsworth to Smallwood 138 KV line  | Reconductor line                            | \$1,083,000   | 8% (\$91,000)         |
| Tanner to Lcec No. 5 138 KV             | Reconductor line                            | \$3,327,000   | 65% (\$2,162,000)     |
|                                         | Total allocated to project                  |               | \$30,524,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.